



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Design and Implementation of a Smart University ERP System with Role-Based Access Control and Academic Resource Management

CSA0904-Programming in Java

Under Supervision of:

Dr.Thalapathi Rajasekaran R

Presented By:

M.Himmath Sai(192411233)

M.Maruthi(192425257)

K.V.S.S.Jeswanth Babu(192472012)

V.Sai Charan(192472159)

P.Venkat Revanth(192311130)

Problem Statement



Data Silos & Security Risks

Manual campus management creates fragmented records and exposes sensitive data to unauthorized access.

No Centralized Access

Students, Faculty, and Admins lack a unified platform, leading to duplicated effort and miscommunication.

Inefficient Academic Operations

Grade processing, attendance tracking, and fee records are handled manually — slow, error-prone, and unsustainable.



Project Scope & Objectives



Secure Java Desktop ERP

Built with Swing/JavaFX and MySQL for a robust, cross-functional academic platform.



Role-Based Access Control

Distinct access privileges for Admin, Faculty, and Student roles enforced at every layer.



Full Academic Automation

Course registration, attendance, exams, fees, library, and reporting — all automated.



Engineer to Excel

Literature Review



S.No	Author name	Year	Title of the paper	DrawBack	Overcome
1	Sharma, R., & Gupta, N.	2025	Role-Based Access Control in Higher Education Information Systems	Difficult management of permissions for large numbers of users.	Dynamic role assignment and centralized permission management.
2	Patel, D., & Mehta, S.	2025	Smart University ERP with Integrated Academic Resource Management	Limited analytics and reporting capabilities.	Include real-time dashboards, academic analytics, and performance reports for better decision-making.
3	Chen, X., & Wang, J.	2024	Intelligent Academic Resource Scheduling Using AI	Scheduling conflicts and inefficient classroom allocation.	Use automated scheduling algorithms to optimize academic resources.
4	Zhang, Y., & Li, H.	2024	Academic Resource Management System Using Web Technologies	Manual resource allocation leads to inefficient utilization.	Automate classroom, laboratory, and library resource allocation.
5	Ahmed, R., & Hussain, S.	2023	Secure Educational Management System Using Authentication Techniques	Weak authentication mechanism vulnerable to unauthorized access.	Apply secure authentication with RBAC and encrypted user credentials.

Technical Background & Tools



Core Language & Frameworks

Java (JDK 17+),
Swing/JavaFX for
GUI



Database & Data Access

MySQL, JDBC API,
Database
Transactions



Programming Concepts

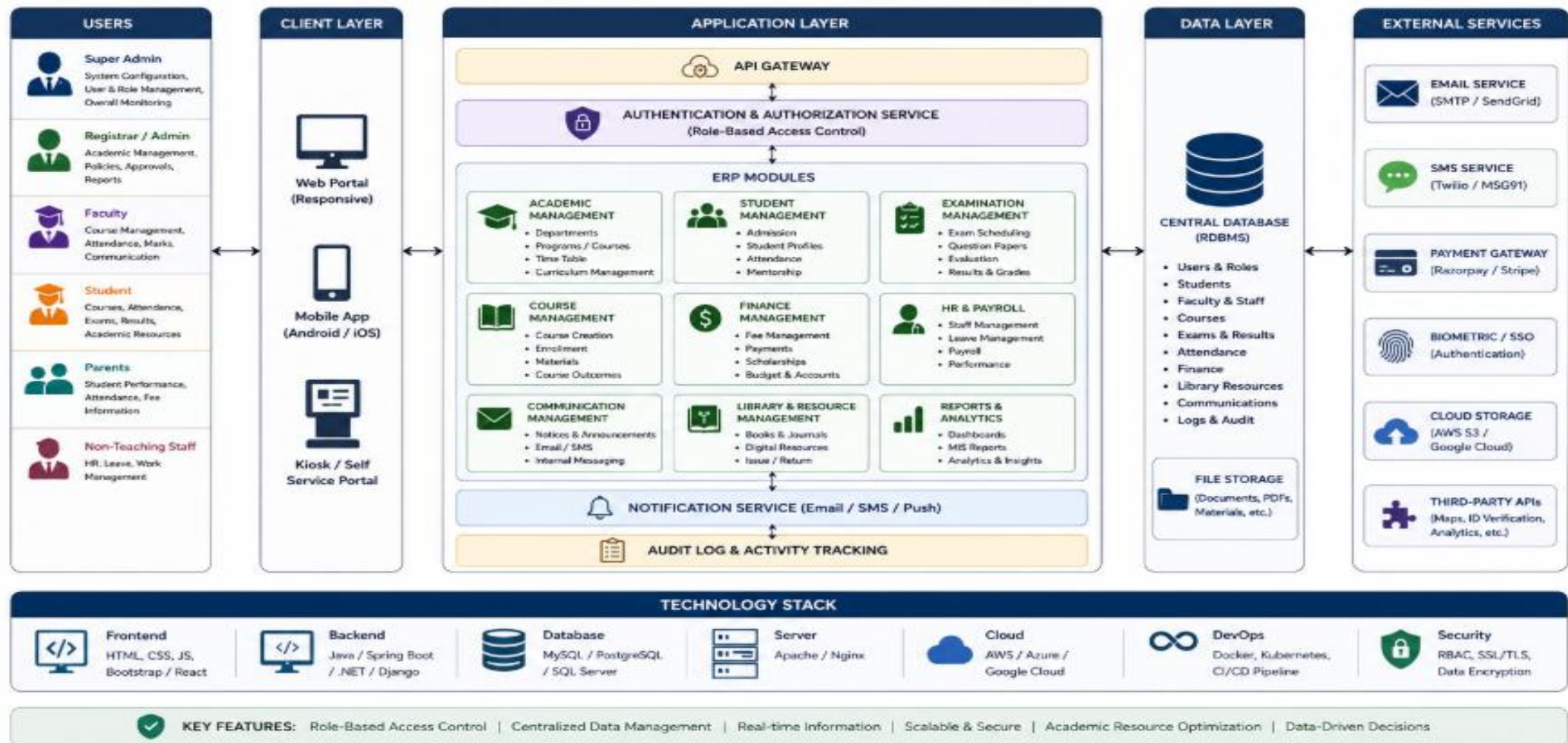
OOP, Collections
Framework, File
Handling, Exception
Handling, BCrypt
Password Hashing





Engineer to Excel

System Architecture



Module 1: Student & Faculty Management

Core User Engine

- ✓ **Centralized User Profiles:** Complete digital registration and profile creation for both students and faculty members.
- ✓ **Dynamic Role Assignment:** Precise mapping of privileges — Students view grades/attendance; Faculty enter marks/schedules.
- ✓ **Academic Lifecycle Records:** Historical logs of enrollment, departmental transfers, section assignments, and personal credentials.
- ✓ **Directory Search & Filters:** Instant administrative query tools to filter records by batch, semester, or major.



Module 2: Course & Exams Management

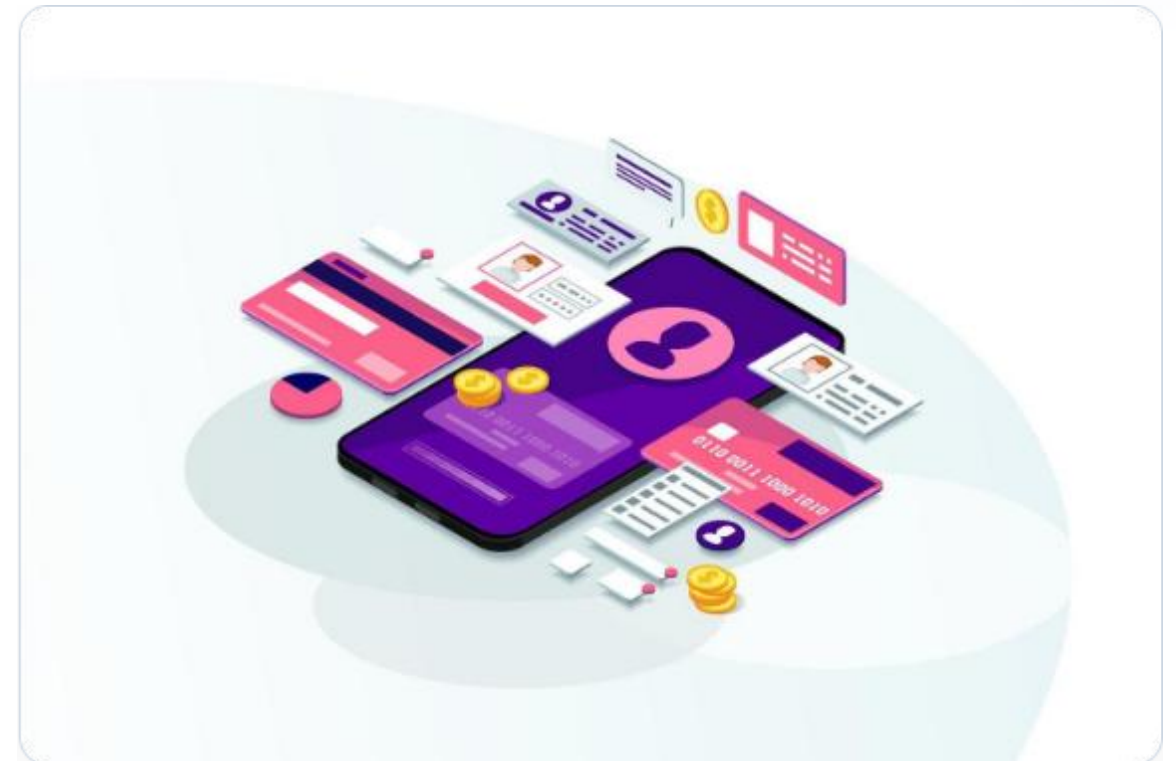
- ✓ **Course Registration:** Automated course offering catalog, prerequisite checking, and student credit tracking.
- ✓ **Real-Time Attendance:** Daily faculty attendance logging with automated threshold warnings (< 75% alerts).
- ✓ **Exam Scheduling & Grading:** Automated GPA/CGPA computation, exam hall mapping, and grade sheet generation.
- ✓ **Classroom Allocation:** Prevents timetable overlaps and optimizes hall capacity utilization.



Module 3: Finance & Resources Management

Asset Engine

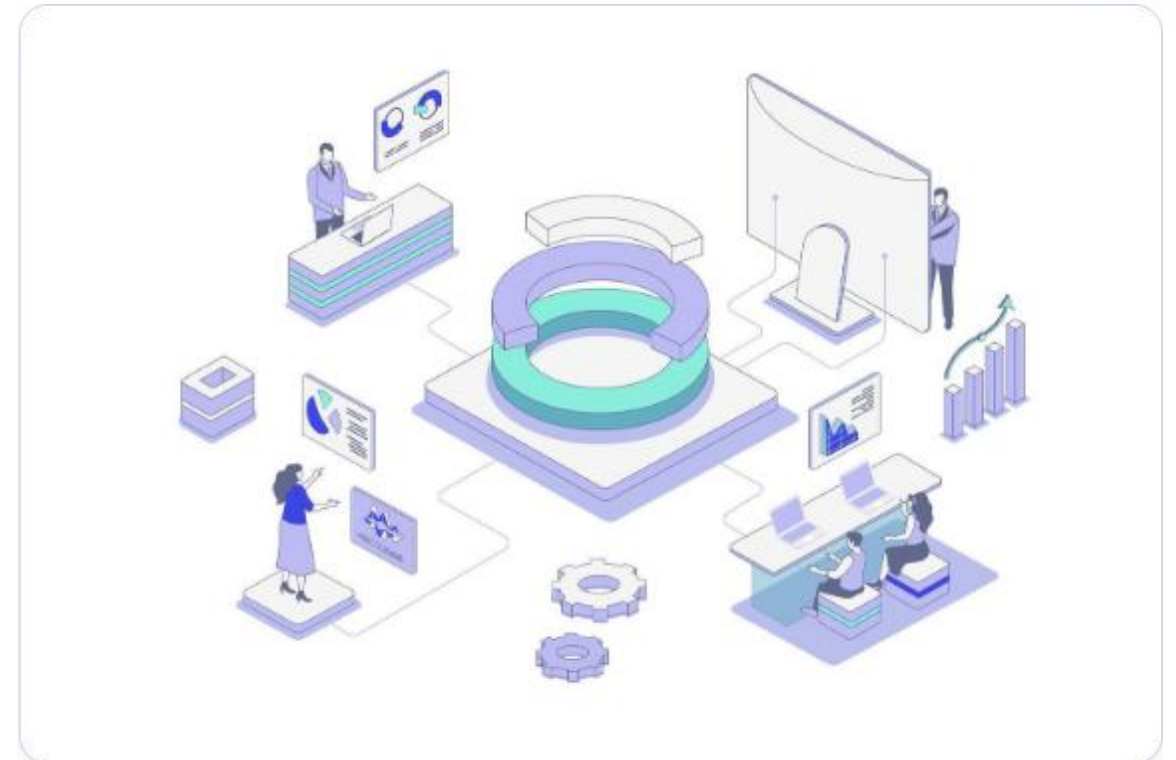
- ✓ **Fee Processing & Invoicing:** Automated tuition, hostel, and exam fee structure generation with pending dues tracking.
- ✓ **Financial Transaction Logs:** Immutable ledger recording all incoming payments and digital receipt issuance.
- ✓ **Library Resource Tracking:** Digital cataloging of books, journals, issue/return dates, and automated overdue fine calculations.
- ✓ **Inventory Management:** Lab equipment and campus infrastructure tracking to streamline administrative requests.



Module 4: Reports & Analytics

Executive Intelligence

- ✓ **Executive Dashboard:** Real-time visual overview of total student count, faculty ratio, fee collection status, and active courses.
- ✓ **MIS Report Generation:** Exportable performance reports in PDF/Excel format for accreditation and university audits.
- ✓ **Academic Trend Analytics:** Comparative pass percentage analysis and department-wise GPA distribution tracking.
- ✓ **Automated Insights:** Identifies at-risk students based on attendance patterns and mid-term evaluations.



Module 5: Security & Audit

- ✓ **BCrypt Credential Encryption:** Passwords hashed with salt before database storage to eliminate plain-text exposure.
- ✓ **Role-Based Authorization Engine:** Every SQL query and UI menu element enforces user role verification at runtime.
- ✓ **Comprehensive Audit Trail:** Timestamps and user IDs logged for every critical update, grade change, or financial edit.
- ✓ **Session & SQL Safety:** Prepared Statements prevent SQL Injection attacks; automatic idle session timeout safeguards terminal access.



Evaluation Metrics & Testing

How We Measure Success

The system is validated across three critical dimensions to ensure it meets academic and enterprise-grade standards before deployment.

- ❏ Zero transaction failures is a hard requirement for the database performance benchmark.



Security

Role authorization verification and BCrypt hashed password validation across all user types.



Database Performance

Query execution speed, JDBC connection pooling efficiency, and zero transaction failures.



System Usability

Functional CRUD operations verified and clean dashboard navigation tested end-to-end.



Expected Outcomes

Centralized Management

A unified academic and administrative management system eliminating data silos across all departments.

Secure Automation

Automated grade generation and attendance tracking with role-based security enforced at every step.

Real-Time Analytics

Efficient real-time analytics dashboards and comprehensive financial transaction logging for administrators.



Engineer to Excel

